

Combinatorial Geometry

Problem 1. 2015 points are given in a plane such that from any five points we can choose two points with distance less than 1 unit. Prove that 504 of the given points lie on a unit disc.

Problem 2. Let there be given a set of 1996 equal circles in the plane, no two of them having common interior points. Prove that there exists a circle touching at most three other circles.

Problem 3. There are k points inside a convex polygon $\mathcal{P}$ with n vertices, where $2 \leq k \leq \left\lfloor \frac{n}{2} \right\rfloor$. Prove that we can choose $2k$ vertices of $\mathcal{P}$ such that the polygon formed by these vertices contains all the k points.

Problem 4. Let $n \geq 2$ be an integer. Find the smallest positive integer m for which the following holds: given n points in the plane, no three on a line, there are m lines such that no line passes through any of the given points, and for all points $X \neq Y$ there is a line with respect to which X and Y lie on opposite sides.

Problem 5. A city is a point on the plane. Suppose there are $n \geq 2$ cities. Suppose that for each city X , there is another city $N(X)$ that is strictly closer to X than all the other cities. The government builds a road connecting each city X and its $N(X)$; no other roads have been built. Suppose we know that, starting from any city, we can reach any other city through a series of road.

We call a city Y *suburban* if it is $N(X)$ for some city X . Show that there are at least $\frac{n-2}{4}$ suburban cities.

Problem 6. Let $\mathcal{S}$ be an infinite set of points on the plane. Suppose for any $A, B, C \in \mathcal{S}$, the distance from A to BC is an integer. Prove that all the points in $\mathcal{S}$ are collinear.

Problem 7. There are 100 circles of radius one in the plane. A triangle formed by the centres of any three given circles has area at most 2017. Prove that there is a line intersecting at least three of the circles.

Problem 8. Twenty ants live on the faces of an icosahedron, one ant on each face, where the icosahedron have each side with length 1. Each ant moves in a counterclockwise direction on each face, along the edges. The speed of each ant must be no less than 1 always. Also, if two ants meet, they should meet at the vertex of the icosahedron. If five ants meet at the same time at a vertex, we call that a *collision*. Can the ants move forever, in a way that no collision occurs?